BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 4

Standard Limits

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1. INTRODUCTION

In Unit 3, we ventured into the core of calculus, delving into the concepts of limits and continuity, which are fundamental to understanding the behavior of functions under various conditions. We started with an intuitive grasp of limits, exploring how functions behave as they approach a specific point, even when they're not explicitly defined at that point. This exploration spanned across different types of functions - polynomial, rational, trigonometric, exponential, and logarithmic - each presenting unique scenarios under the lens of limits.

Continuity took this understanding further, requiring a function to be seamless, without abrupt jumps, at a given point. The notion that a function is continuous if its limit at a point equals the function's value at that point bridges the gap between abstract mathematical concepts and their tangible applications in the real world. Whether it's the smooth flow of electricity or the predictable path of a moving object, continuity in functions ensures predictability and stability in numerous physical phenomena and engineering applications.

As we transition into Unit 4, the spotlight turns to derivatives, a pivotal concept in calculus that represents the rate of change of a function with respect to its independent variable. For students diving into computer applications, understanding derivatives is not just academic; it's a tool that equips you to model and solve real-world problems where change is a constant. From optimizing algorithms to understanding the dynamics of systems and beyond, derivatives offer a lens through which we can quantify and manipulate the concept of change.

The derivative is defined using the concept of limits, capturing the essence of change at an infinitesimally small scale. This definition paves the way for understanding more complex phenomena in physics, engineering, and economics, where the rate of change is crucial for predictions, optimizations, and understanding system behaviors.

Navigating through the complex landscape of limits, continuity, and derivatives demands a strategic approach to learning. It starts with building a solid foundation in the basic concepts, followed by gradual exposure to more complex scenarios and problem-solving exercises. Visual aids like graphs and diagrams play a crucial role in demystifying abstract concepts, making them more accessible and understandable.

Engaging with practical examples that apply these mathematical concepts to real-world scenarios can bridge the gap between theory and practice. Collaboration with peers and instructors, alongside leveraging online resources and simulations, can provide diverse perspectives and deepen understanding.

1.1 Objectives

At the end of this unit, you will be able to,

- List the fundamental concepts of limits and continuity in calculus.
- Explain the relationship between the concept of a limit and the definition of continuity in a function.
- Calculate the limit of simple polynomial functions using basic limit laws.
- Explain the continuity of a given function at a specific point by examining its limit and function value at that point

2. DEFINITION OF DERIVATIVE USING THE LIMITS

The derivative of a function can be defined using the concept of a limit. The derivative of a function, denoted as f'(x), represents the rate of change of the function with respect to its independent variable, usually denoted as 'x'. To define the derivative using limits, we consider the following expression:

$$f'(x) = \lim_{h \to 0} \frac{[f(x+h) - f(x)]}{h}$$

Where,

f'(x): represents the derivative of the function f(x) at the point x.

$\lim_{h \to 0}$: This is a limit notation, indicating that we are taking the limit as the change in 'x', denoted as 'h', approaches zero. In other words, we are looking at the behavior of the function as the interval over which we calculate the rate of change becomes infinitesimally small.

[f(x + h) - f(x)]: This calculates the difference in function values between two points: one at 'x' and another at 'x + h'. This difference represents the change in the function's value as 'x' is disturbed by a small amount 'h'.

/ h: We divide the above difference by 'h', which represents the small change in 'x'. This step normalizes the rate of change, making it independent of the specific size of 'h'.

In essence, the derivative of a function at a certain point 'x' is formally defined as the limit of the average rate of change of the function as the period over which the change is being examined approaches zero. Put simply, it measures the rate at which the value of a function changes when there is a little variation in the independent variable 'x', hence offering an indication of the function's slope at that specific moment.

The derivative of a function is a fundamental concept in calculus, allowing us to understand how a function's output changes as its input varies. It provides the instantaneous rate of change of the function at a specific point. This concept is particularly important when dealing with nonlinear functions, where the rate of change isn't constant.

The core idea behind the derivative using limits is to capture this instantaneous rate of change by considering smaller and smaller intervals around a point. Here's a step-by-step explanation of the formula:

1. We start with a function f(x), and we're interested in finding its derivative at a specific point 'x'. The derivative, denoted as f'(x), tells us how much the function changes as 'x' changes, but we want to know this change at an exact point, not over a large interval.

2. To do this, we introduce a small change in the input, represented by 'h'. We calculate the change in the function's output by evaluating f(x + h) - f(x). Essentially, we're measuring the function's difference between 'x + h' and 'x'. This difference reflects how the function values change as we move from 'x' to 'x + h'.

3. Next, we divide this difference by 'h'. This division normalizes the rate of change because 'h' can be made smaller and smaller, almost like zooming in on the point 'x'. The result is a value that represents the rate of change as 'h' approaches zero.

4. Finally, we take the limit as 'h' approaches zero, denoted as lim(h→0). This process captures the instantaneous rate of change. As 'h' gets smaller and smaller, the value approaches the actual derivative at 'x', and that's what we call f'(x).

Lets look at few examples to see the derivatives of different functions

Derivative of a Linear Function:

Let's consider the function f(x) = 2x.

To find its derivative at a specific point, say x = 3, we apply the limit definition:

$$f'(3) = \lim_{h\to 0} \frac{2(3+h) - 2(3)}{h}$$

$$= \lim_{h\to 0} \frac{2h}{h}$$

$$= \lim_{h\to 0} 2 = 2$$

So, the derivative of f(x) = 2x at x = 3 is 2.

Derivative of a Quadratic Function:

Consider the function g(x) = x^2.

To find its derivative at a specific point, say x = 2, we use the limit definition:

$$g'(2) = \lim_{h\to 0}\frac{(2+h)^2 - 2^2}{h}$$

$$= \lim_{h\to 0}\frac{4h+h^2}{h}]$$

$$= \lim_{h\to 0} 4 + h$$

$$= 4$$

The derivative of g(x) = x^2 at x = 2 is 4.

Derivative of an Absolute Value Function:

Let's explore the function h(x) = |x|. To find its derivative at x = 1, we apply the limit definition:

$$h'(1) = \lim_{h\to 0}\frac{|1+h| - |1|}{h}$$

$$= \lim_{h\to 0}\frac{|1+h| - 1}{h}$$

As h approaches zero from both sides, we consider the behaviour of |1 + h|.

When h is positive, |1 + h| is 1 + h, and when h is negative, |1 + h| is -(1 + h). So:

$$h'(1) = \lim_{h\to 0}\frac{1+h-1}{h}$$

$$= \lim_{h\to 0}\frac{h}{h}$$

$$= 1$$

The derivative of h(x) = |x| at x = 1 is 1.

Derivative of a Piecewise Function:

Suppose we have the piecewise function k(x) defined as follows:

$$k(x) = \{ x^2, \text{ if } x > 0 \ 2x, \text{ if } x \leq 0 \}$$

To find the derivative at x = 0, we calculate it separately for the two cases:

For x > 0:

$$k'(0) = \lim_{h \to 0} \frac{(0+h)^2 - 0^2}{h}$$

$$= 0$$

For x ≤ 0:

$$k'(0) = \lim_{h \to 0} \frac{2(0+h) - 2(0)}{h}$$

$$= \lim_{h \to 0} \frac{2h}{h}$$

$$= 2$$

So, the derivative of k(x) at x = 0 is 0 from the left side and 2 from the right side. This indicates that the derivative is not defined at x = 0 for this piecewise function.

2.1 Derivative by First Principle

The derivative of a function represents the rate at which that function is changing at a specific point. To define the derivative using limits or the first principle, we look at the instantaneous rate of change of a function at a single point. The derivative of a function f(x) at a particular point 'a' is denoted as f'(a).

The derivative f'(a) can be defined using the following limit-based formula:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Where:

- **f'(a)**: The derivative of the function f(x) at the point 'a'.

- $\lim_{h \to 0}$: This is a limit as the "change" in the x-value (h) approaches zero. This limit process makes the derivative an instantaneous rate of change.
- **f(a + h)**: The value of the function f(x) at a point a + h, which is slightly to the right of 'a'.
- **f(a)**: The value of the function f(x) at the point 'a'.
- **h**: A small change in the x-value. We take the limit as h approaches zero, making this change infinitesimally small.

The limit expression [f(a + h) - f(a)] / h calculates the average rate of change of the function f(x) as we move from 'a' to 'a + h'. As h approaches zero, this average rate of change becomes the instantaneous rate of change at 'a', which is the derivative.

This definition is fundamental to the concept of the derivative in calculus. It allows us to find the slope of the tangent line to a curve at a specific point, indicating how the function behaves precisely at that location.

To calculate the derivative using the first principle, you typically simplify the limit expression by applying algebraic operations and cancelling terms that become zero as h approaches zero. The result gives you the derivative of the function at the chosen point 'a'.

Example 1: Find the derivative of the function f(x) = 2x at x = 3.

Solution:

1. Start with the definition of the derivative:

$$f'(3)=\lim_{h \to 0} \frac{f(3+h)-f(3)}{h}$$

2. Substitute the function f(x) = 2x:

$$f'(3)=\lim_{h \to 0} \frac{2(3+h)-2(3)}{h}$$

3. Simplify the expression:

$$f'(3)= \lim_{h \to 0} \frac{6+2h-6}{h}$$

4. Cancel out the common terms:

$$f'(3)= \lim_{h\to 0} \frac{2h}{h}$$

5. Cancel the 'h' in the numerator and denominator:

$$f'(3)= \lim_{h\to 0} 2$$

6. Now, take the limit as h approaches 0:

$$f'(3)=2$$

So, the derivative of the function f(x) = 2x at x = 3 is 2.

Example 2: Find the derivative of the function f(x) = x^2 at x = 4.

Solution:

1. Start with the definition of the derivative:

$$f'(4)= \lim_{h\to 0} \frac{f(4+h)-f(4)}{h}$$

2. Substitute the function f(x) = x^2:

$$f'(4)=\lim_{h\to 0} \frac{(4+h)^2-4^2}{h}$$

3. Expand and simplify:

$$f'(4)= \lim_{h\to 0} \frac{16+8h+h^2-16}{h}$$

4. Cancel out the common terms:

$$f'(4)= \lim_{h\to 0} \frac{8h+h^2}{h}$$

5. Factor out an h from the numerator:

$$f'(4)= \lim_{h\to 0} \frac{h(8+h)}{h}$$

6. Cancel the 'h' in the numerator and denominator:

$$f'(4)= \lim_{h\to 0} (8+h)$$

7. Now, take the limit as h approaches 0:

$$f'(4)=8+0=8$$

So, the derivative of the function f(x) = x^2at x = 4 is 8.

These examples illustrate how to find derivatives using the definition of a derivative

3. GENERAL DERIVATIVE FORMULAE'S

1. **Power Rule:**

Formula:

If $f(x)=x^n$, where n is a constant, then the derivative $f'(x)$ is given by:

$$f'(x)=n\cdot x(n-1)$$

The power rule states that when you have a term in the form xn, the derivative is n times the term x(n-1). For example, if you have f(x)=x3, its derivative is $f'(x)=3x2$.

2. **Constant Rule:**

Formula:

If f(x)=c, where c is a constant, then the derivative $f'(x)$ is always zero.

The constant rule tells us that the derivative of a constant is always zero because a constant value does not change as x varies.

3. **Sum/Difference Rule:**

Formula:

If $f(x)=g(x)\pm h(x)$, then the derivative $f'(x)$ is the sum (or difference) of the derivatives of g(x) and h(x), i.e.,

$$f'(x)=g'(x)\pm h'(x).$$

This rule allows us to find the derivative of a function that is a sum or difference of two functions by simply taking the derivative of each function separately.

4. **Product Rule:**

Formula:

If $f(x)=g(x)\cdot h(x)$, then the derivative $f'(x)$ is given by:

$$f'(x)=g(x)\cdot h'(x)+g'(x)\cdot h(x)$$

The product rule is used to find the derivative of a product of two functions. It involves taking the derivative of one function while keeping the other constant, then adding the derivative of the second function while keeping the first one constant.

5. **Quotient Rule:**

Formula:

If $f(x)=\frac{g(x)}{h(x)}$ then the derivative $f'(x)$ is given by:

$$f'(x)=\frac{g'(x)\cdot h(x)-g(x)\cdot h'(x)}{(h(x))^2}$$

The quotient rule is used to find the derivative of a quotient of two functions. It involves taking the derivative of the numerator, multiplying it by the denominator, and then subtracting the product of the numerator and the derivative of the denominator, all divided by the square of the denominator.

Example:

Find the derivative of $f(x)=4x^3-2x^2+7x-1$.

Solution:

1. Apply the power rule to each term:
   - For the term $4x^3$, the derivative is $12x^2$ (using the power rule with n=3).
   - For the term $-2x^2$, the derivative is $-4x$ (using the power rule with n=2).
   - For the term $7x$, the derivative is 7 (using the power rule with n=1).
   - For the constant term -1, the derivative is 0 (since the derivative of a constant is always zero).
2. Combine the derivatives of individual terms:

   $f'(x)=12x^2-4x+7$

   So, the derivative of $f(x)$ is $f'(x)=12x^2-4x+7$.

Example:

Find the derivative of $h(x)=3x^2-2x+5$.

Solution:

1. Apply the sum/difference rule to each term:
   - For the term $3x^2$, the derivative is $6x$ (using the power rule with $n=2$).
   - For the term $-2x$, the derivative is -2 (using the power rule with $n=1$).
   - For the constant term 5, the derivative is 0 (since the derivative of a constant is always zero).
2. Combine the derivatives of individual terms: $h'(x)=6x-2$

   So, the derivative of $h(x)$ is $h'(x)=6x-2$.

Example:

Find the derivative of the constant function $f(x)=5$.

Solution:

1. Apply the constant rule, which states that the derivative of a constant is always zero. $f'(x)=0$

   So, the derivative of the constant function $f(x)=5$ is $f'(x)=0$.

Example:

Find the derivative of $g(x)=x^2 \cdot e^x$

Solution:

1. Apply the product rule, which states that if $f(x)=g(x) \cdot h(x)$, then the derivative $f'(x)$ is given by: $f'(x)=g(x) \cdot h'(x)+g'(x) \cdot h(x)$
2. Identify $g(x)$ and $h(x)$:
   - $2g(x)=x^2$
   - $h(x)=e^x$
3. Find the derivatives of $g(x)$ and $h(x)$:
   - $g'(x)=2x$ (using the power rule for x^2)

- $h'(x)=e^x$ (the derivative of e^x is e^x)

4. Apply the product rule formula:

$$g(x)\cdot h'(x)+g'(x)\cdot h(x)=x^2\cdot e^x+2x\cdot e^x$$

So, the derivative of $g(x)=x^2\cdot e^x$ is $g'(x)=x^2\cdot e^x+2x\cdot e^x$.

Example:

Find the derivative of $h(x)=x^2/e^x$.

Solution:

1. Apply the quotient rule, which states that if $f(x)=g(x)/h(x)$, then the derivative $f'(x)$ is given by:

$$f'(x)=\frac{g'(x)\cdot h(x)-g(x)\cdot h'(x)}{(h(x))^2}$$

2. Identify $g(x)$ and $(h(x)$:
   - $g(x)=x^2$
   - $h(x)=e^x$
3. Find the derivatives of $g(x)$ and $(h(x)$:
   - $g'(x)=2x$ (using the power rule for x^2)
   - $h'(x)=e^x$ (the derivative of e^x is e^x)
4. Apply the quotient rule formula:

$$f'(x)=\frac{(2x\cdot e^x-x^2\cdot e^x)}{(e^x)^2}$$

$$=\frac{2xe^x-x^2e^x}{e^{2x}}$$

So, the derivative of $h(x)=x^2/e^x$ is

$$h'(x)=\frac{2xe^x-x^2e^x}{e^{2x}}$$

These examples demonstrate the application of the constant rule, product rule, and quotient rule for finding derivatives of different functions. These rules are essential tools in calculus for solving more complex problems involving various functions.

4. INDETERMINATE FORMS

Indeterminate forms are specific mathematical expressions that cannot be easily evaluated without further analysis or manipulation. These forms often arise when trying to find limits in calculus or when working with complex mathematical expressions. Indeterminate forms occur because the value of the expression is uncertain, even though it might seem that it should be a straightforward calculation. The most common indeterminate forms are "0/0," "∞/∞," and "∞ - ∞."

1. **0/0 (Zero over Zero): This** form arises when both the numerator and denominator approach zero as they get closer to a specific point, making it unclear what the limit is without simplification or further analysis.
2. **∞/∞ (Infinity over Infinity):** In this case, both the numerator and denominator grow without bound as they approach a particular point, and the limit is uncertain.
3. **∞ - ∞ (Infinity minus Infinity):** Here, you have two quantities that tend toward infinity, but their difference is uncertain.

Examples:

- Consider the limit $\lim_{x\to 1} x^2-1/x-1$. As x approaches 1, both the numerator and denominator become zero, resulting in the indeterminate form 0/0.
- For $\lim_{x\to\infty} x^3/e^x$, as x grows large, both the numerator and denominator become very large, leading to the indeterminate form ∞/∞.

Resolving indeterminate forms typically involves simplifying the expression or applying limit theorems. Techniques such as L'Hôpital's Rule, factoring, and algebraic manipulation are used to evaluate these limits effectively.

- L'Hôpital's Rule is often applied when the limit results in 0/0 or ∞/∞. It allows you to find the limit of the ratio of the derivatives of the numerator and denominator.
- Factoring and algebraic manipulation can help simplify the expression to a form that can be directly evaluated.

Indeterminate forms are frequently encountered in calculus, especially when dealing with limits, derivatives, and integrals. They are crucial for solving real-world problems in fields

like physics, engineering, economics, and more. By resolving these forms, mathematicians and scientists gain insights into the behaviour of functions and systems.

Understanding indeterminate forms is essential for mastering calculus and applying mathematical principles to practical scenarios. These forms highlight the importance of careful analysis and mathematical techniques in dealing with complex mathematical situations

1. Limit of 0/00/0

When you encounter the indeterminate form 0/00/0, it means that as the variable (usually denoted as x) approaches a certain value, both the numerator and denominator of a function approach zero. This form arises when you have a point of discontinuity or a removable singularity in a function. To evaluate a limit in this form, you can use various techniques, including factoring, simplification, or L'Hôpital's Rule.

Example:

Consider the limit $\lim x \to x^2 - 4/x-2$. As x approaches 2, both the numerator and denominator approach zero. You can factor the expression as $(x-2)(x+2)/x-2$, cancel the common factor, and evaluate the limit as $\lim x \to 2\ (x+2)=4$. So, the limit is 4.

2. Limit of ∞/∞

The indeterminate form ∞/∞ arises when both the numerator and denominator of a function grow without bound as a variable approaches a particular value. This form indicates an unclear result, as the relative rates at which the numerator and denominator grow can vary. To evaluate a limit in this form, you typically apply techniques such as factoring, rationalizing, or L'Hôpital's Rule.

Example:

Consider the limit $\lim x \to \infty\ x^2+3x/2x^3-7x$. As x becomes large, both the numerator and denominator grow without bound. You can use polynomial division to simplify the expression and then evaluate the limit. The result might be a finite number or infinity, depending on the specific function.

3. Limit of ∞−∞

The indeterminate form ∞−∞ indicates a situation where two quantities grow without bound, but their difference is uncertain. This form is often encountered when dealing with limits that involve subtraction. Resolving this indeterminate form requires algebraic manipulation or other limit techniques.

Example:

Consider the limit $\lim_{x\to 0} x^2-x^3$. As x approaches 0, both terms approach zero, but their difference is uncertain. To evaluate this limit, you can factor the expression, leading to $\lim_{x\to 0} x^2(1-x)$. The limit evaluates to 0 since both factors tend to zero.

In all of these cases, understanding the specific context and applying appropriate techniques is essential to resolve the indeterminate forms and evaluate the limits accurately. These techniques are fundamental tools in calculus for understanding the behaviour of functions and their limits.

5. SUMMARY

So, we've just wrapped up this whirlwind journey through the realms of limits, continuity, and derivatives in our calculus adventure. It's been quite the ride, hasn't it? Let's take a moment to chat about the ground we've covered and what it all means, especially for those of us venturing into the world of computer applications.

We kicked things off with limits, this fascinating concept that essentially asks, "What's the function heading towards as we approach a certain point?" It's like trying to predict where a thrown ball will land as it gets closer and closer to the ground. We saw how this plays out in various types of functions, from the straightforward polynomial to the more complex trigonometric and logarithmic functions. It's all about understanding behavior at the brink of change, a skill crucial not just in math but in solving real-world problems where predicting outcomes is key.

Then we delved into continuity, a concept that ensures smooth sailing without any unexpected jumps or breaks. It's akin to ensuring a seamless music playlist, where each song transitions smoothly into the next without abrupt stops. This idea of uninterrupted flow is

crucial in designing algorithms and systems that are reliable and predictable, ensuring they perform as expected under various conditions.

Transitioning into derivatives was like stepping into a new dimension, where we zoom in on the rate of change at an infinitesimally small scale. It's the mathematical equivalent of examining a high-speed video frame by frame to catch every subtle movement. For us in computer applications, it's about understanding how changes in input affect the output, a fundamental aspect of optimizing algorithms, modeling systems, and even in the graphical rendering in video games and simulations.

The beauty of derivatives lies in their definition through limits. It brings everything full circle, showing us that understanding the essence of change at a point gives us powerful tools to model and manipulate the digital world around us. It's not just about numbers and equations; it's about capturing the essence of change, movement, and progression.

So, as we close this unit, remember that the journey through limits, continuity, and derivatives isn't just an academic endeavor. It's a toolkit for the real world, equipping us with the means to predict, analyze, and innovate. Whether it's optimizing data flow, smoothing out animations, or predicting market trends, these concepts are the bedrock upon which we build and understand the ever-evolving digital landscape. Here's to all the future applications we'll explore and the innovations we'll create with these powerful mathematical tools at our disposal!

INSPIRED BY LIFE

6. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What is the definition of a derivative using the limits (Derivative by first principle)?
   a) $f'(a)=\lim_{h\to a} f(a)-f(a-h)/h$
   b) $f'(a)=\lim_{h\to 0} f(a+h)-f(a)/h$
   c) $f'(a)=\lim_{h\to 0} f(a)-f(a+h)/h$
   d) $f'(a)=\lim_{h\to a} f(a+h)-f(a)/h$
2. If you want to find the derivative of a function at a specific point using the first principle, what value does 'h' approach as it gets smaller?
   a) The value of 'h' approaches zero.
   b) The value of 'h' approaches infinity.
   c) The value of 'h' approaches one.
   d) The value of 'h' approaches 'a,' the point of interest.
3. Given the function f(x) = 3x^3 - 2x^2 + 5x - 1, what is the derivative of this function at x = 2 using the first principle?
   a) 23
   b) 31
   c) 35
   d) 41
4. If you have a function f(x) and you want to find the derivative at a specific point, what does the first principle of differentiation involve?
   a) Evaluating the function at that point and dividing by 'a.'
   b) Taking the limit as 'h' approaches zero of the difference quotient involving the function.
   c) Finding the slope of the tangent line at that point.
   d) Calculating the area under the curve between two points.

5. What is the derivative of $f(x)=4x^3-2x^2+7x-1$ using the power rule?
   a) $f'(x)=3x^3-4x^2+7$
   b) $f'(x)=12x^2-4x+7$
   c) $f'(x)=12x^2-4x$
   d) $f'(x)=4x^3-4x^2+7$
6. Which rule is used to find the derivative of a product of two functions?
   a) Power Rule
   b) Sum/Difference Rule
   c) Product Rule
   d) Quotient Rule
7. If you encounter an expression of the form $\infty - \infty$, which general derivative formula can be helpful in simplifying it?
   a) Power Rule
   b) Sum/Difference Rule
   c) Product Rule
   d) Quotient Rule
8. Which rule is often applied when you have an expression in the form $0/0$ or ∞/∞ and want to evaluate the limit?
   a) Power Rule
   b) Sum/Difference Rule
   c) L'Hôpital's Rule
   d) Chain Rule
9. What type of indeterminate form is encountered when both the numerator and the denominator of a fraction approach zero as the limit is evaluated?
   a) $0/0$
   b) $1/0$
   c) ∞/∞
   d) $0/1$

10. When evaluating a limit, if both the numerator and the denominator grow without bound as they approach a particular point, which indeterminate form is encountered?
    a) 0/0
    b) 1/∞
    c) ∞/∞
    d) 0/1
11. In which indeterminate form are there two quantities that become infinitely large, but their difference is uncertain?
    a) 0/0
    b) ∞/0
    c) ∞/∞
    d) ∞ - ∞
12. Which mathematical technique is often used to evaluate indeterminate forms of 0/0 or ∞/∞?
    a) Factoring
    b) Integration
    c) L'Hôpital's Rule
    d) Differentiation

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7. TERMINAL QUESTIONS

1. What is the fundamental concept behind the derivative when using the first principle?
2. Find the derivative of $f(x)=3x^2-2x+1$ using the first principle at the point $x=2$.
3. Apply the power rule to find the derivative of $g(x)=4x^3-2x^2+7x-1$.
4. What is the sum rule for derivatives, and how is it used to find the derivative of a sum of functions?
5. Explain the concept of indeterminate forms in the context of evaluating limits. Provide an example of an indeterminate form and how it is resolved.
6. When evaluating a limit and encountering the $\infty - \infty$ indeterminate form, what mathematical techniques can be used to simplify and resolve it?
7. Provide an example of a limit that results in the $0/0$ indeterminate form. Explain how you would resolve this limit.
8. How is L'Hôpital's Rule applied to evaluate limits involving the ∞/∞ indeterminate form?
9. Explain the concept of indeterminate forms and why they are essential in the study of calculus. Provide a real-world example where resolving an indeterminate form is crucial.
10. Define the $\infty - \infty$ indeterminate form. Provide an example of a limit that results in this form and explain how to resolve it.

8. ANSWERS

Self-assessment Answers:

1. B) $f'(a)=\lim_{h\to 0} f(a+h)-f(a)/h$
2. A) The value of 'h' approaches zero.
3. B) 31
4. B) Taking the limit as 'h' approaches zero of the difference quotient involving the function
5. B) $f'(x)=12x^2-4x+7$
6. C) Product Rule
7. B) Sum/Difference Rule
8. C) L'Hôpital's Rule
9. A) 0/0
10. C) ∞/∞
11. D) ∞ - ∞
12. C) L'Hôpital's Rule

Terminal Question Answers

1. The derivative represents the instantaneous rate of change of a function at a specific point and is defined as the limit of the difference quotient as h approaches 0.
2. $f'(2)=10$
3. $g'(x)=12x^2-4x+7$
4. The sum rule states that the derivative of the sum of two functions is the sum of their derivatives. For functions u(x) and v(x), $(u+v)'=u'+v'$.
5. Indeterminate forms are expressions that cannot be easily evaluated without further analysis. For example, in the limit $\lim_{x\to 1} x^2-1/x-1$, we encounter the 0/0 indeterminate form. This form is resolved by algebraic manipulation or factoring.
6. The ∞ - ∞ indeterminate form can be resolved using algebraic manipulation, factoring, or by employing series expansion methods.
7. Example: $\lim_{x\to 2} x^2-4/x-2$. To resolve this limit, we can factor the expression and cancel common terms.
8. L'Hôpital's Rule allows you to find the limit of the ratio of the derivatives of the numerator and denominator when you have the ∞/∞ indeterminate form.

9. Indeterminate forms are critical in calculus as they represent situations where the limit is uncertain, requiring further analysis. An example is finding the limit of a function representing the velocity of a moving object, which can help predict its position at a particular time.
10. The ∞ - ∞ indeterminate form occurs when you have two quantities that become infinitely large, but their difference is uncertain. Example: $\lim_{x\to\infty} (x^2-x)$. To resolve it, algebraic manipulation is used to simplify the expression.